

Optimality Theory: The good and the bad

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```
source(here::here("_defaults.R"))
```

```
Loading required package: sysfonts
```

```
Loading required package: showtextdb
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ lubridate 1.9.5     ✓ tibble    3.3.1
✓ purrr     1.2.2     ✓ tidyr     1.3.2
```

```
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
library(tinytable)
```

Typology and Conspiracies

Two strengths of OT is how it can account for

- Typological patterns (cross-linguistic comparisons)
- "Conspiracies" within languages.

Typology

There are many asymmetries in syllable structure patterns cross-linguistically



Codas

- Some languages disallow codas altogether
  - CV, *CVC
- Some languages allow codas
  - CV, CVC
- No language *requires* a coda
  - *CV, CVC

Typology

Maybe there's a NoCoda markedness constraint

Languages that Disallow Codas

CVC	NoCoda	Max	Dep
CVC	*!		
☞ CV		*	

CV	NoCoda	Max	Dep
CVC			*!
☞ CV			

Typology

Maybe there's a NoCoda markedness constraint

Languages that Allow Codas

CVC	Max	NoCoda	Dep
☞ CVC		*	
CV	*!		

CV	Max	NoCoda	Dep
CVC			*!
☞ CV			

Typology

- There's no way to rank NoCoda, Max and Dep in a way that winds up *requiring* a coda.
- There's also no language that requires a coda.



Typology

Onsets

- Some languages require syllables to have onsets
  - CV, CVC, *VC, *V
- Some languages allow onsetless syllables
  - CV, CVC, VC, V
- No language *disallows* onsets
  - *CV, *CVC, VC, V

Typology

Maybe there's an Onset markedness constraint

Languages that require onsets

CVC	Onset	Max	Dep
☞ CVC			
VC	*!	*	

VC	Onset	Max	Dep
☞ CVC			*
VC	*!		

Typology

Maybe there's an Onset markedness constraint

Languages that don't require onsets

CVC	Max	Dep	Onset
☞ CVC			
VC	*!		*

VC	Max	Dep	Onset
CVC		*!	
☞ VC			*

Typology

- There's no way to rank Onset, Max and Dep in a way that *disallows* onsets.
- There's also no language that disallows onsets.



Typology

Other typological asymmetries:

Complex Onsets & Codas

- Some languages disallow complex onsets and/or codas
- Some languages allow complex onsets and/or codas
- No language *requires* complex onsets and/or codas

Word Final Devoicing

- Some languages disallow voiced obstruents at the end of words.
- Some languages allow voiced obstruents at the end of words.
- No language *requires* voiced obstruents at the end of words.

Conspiracy

Sometimes, languages seem to do multiple different things to avoid a single kind of outcome.

Conspiracy

Japanese doesn't like having the sequence [tu]

```
tribble(
  ~UR, ~SR,
  "kɑtu", "kɑt͡su",
  "tunɑmi", "t͡sunɑmi"
) |>
  tt()
```

UR	SR
kɑtu	kɑt͡su
tunɑmi	t͡sunɑmi

Conspiracy

tunɑmi	MAX	*TU	IDENT-VOWEL	IDENT-CONT
tunɑmi		*!		
☞ t͡sunɑmi				*
unɑmi	*!			
tonɑmi			*!	

Conspiracy

When borrowing words, Japanese tends to epenthesize [u] to conform to its CV syllable structure.

```
tribble(
  ~UR, ~SR,
  "besbɔl", "besubɑru"
) |>
  tt()
```

UR	SR
besbɔl	besubɑru

Conspiracy

besbɔl	Max	CV	Dep	*O	*U
besbɑr		*!			
beba	*!				
☞ besubaru			**		**
besobaro			**	**!	

Conspiracy

But, when we need to epenthesize a vowel to break up a cluster with [t], Japanese uses [o].

```
tribble(
  ~UR, ~SR,
  "stɪaik", "sut<b>o</b>raiku"
) |>
  tt()
```

UR	SR
stɪaik	sutoraiku

Conspiracy

Both of these look like a “conspiracy” to avoid [tu].

- Assibilate [tu] to [t͡su] to avoid the [tu] sequence.
- Epenthesize [u] to break up clusters, unless that would create the [tu] sequence, in which case use [o]

Conspiracy

straik	MAX	*TU	CV	IDENT-VOWEL	DEP	IDENT-CONT	*O	*U
straik			***					
sai	***!							
suturaiku		*!			***		***	
sut͡suraiku					***	*!		***
☞ sutoraiku					***		*	***
sotoraiku					***		***!	*

Conspiracy

besbɔl	MAX	*TU	CV	IDENT-VOWEL	DEP	*O	*U	IDENT-CONT
besbɑr			*!					
beba	**!							
☞ besubaru					**		**	
besobaro					**	**!		

Conspiracy

tunɑmi	MAX	*TU	CV	IDENT-VOWEL	DEP	IDENT-CONT	*O	*U
tunɑmi		*!						*
tonɑmi				*!			*	
☞ t͡sunɑmi						*		*
unɑmi	*!					*		*

Shortcomings: Opacity

Let's cast our mind back to the Bosnian-Serbian-Croatian Data

```
tribble(
  ~a, ~b, ~c, ~d, ~e,
  "UR", "ledn-i", "ledn", "bel", "okrugl",
  "Epenthesis", "--", "ledan", "--", "okrugal",
  "L-Vocalization", "---", "---", "beo", "okrugao",
  "SR", "ledni", "ledan", "beo", "okrugao"
) |>
  tt(
    colnames = F
  ) |>
  style_tt(
    j = 5,
    bold = TRUE
  ) |>
  style_tt(
    j = 1,
    smallcap = T
  )
```

UR	ledn-i	ledn	bel	**okrugl**
Epenthesis	–	ledan	–	**okrugal**
L-Vocalization	—	—	beo	**okrugao**
SR	ledni	ledan	beo	**okrugao**

Shorcomings: Opacity

We can come up with constraints and constraint rankings to give us each of the transparent forms, but we can't capture the opaque form.

Shortcomings: Opacity

ledn	*Obs-Son#	*L#	Max	Dep	Ident-L
ledn	*!				
☞ ledan				*	
led			*!		

Shortcomings: Opacity

bel	*Obs-Son#	*L#	Max	Dep	Ident-L
bel		*!			
bela				*!	
be			*!		
☞ beo					*

Shortomings: Opacity

The wrong candidate wins

okrugl	*Obs-Son#	*L#	Max	Dep	Ident-L
okrugl	*!	*			
okrugal		*!		*	
okrug			*!		
☞ okrugo					*
okrugao				*!	*